preserves $\hat{\rho}(\xi) = \exp(-\beta\mu\xi^2/2)$. However, this is obvious from our earlier investigations of Ornstein-Uhlenbeck equations.

In order to show that this thermostat is ergodic, we need to demonstrate that the assumptions presented in Sect. 6.4.4 are valid. Assumption 1(ii) requires us to verify that a Hörmander condition (see Definition 6.1) holds for the Nosé-Hoover-Langevin system. Because of the complexity of the high order commutators, we will work here with the assumption of a quadratic potential in the physical model. In some sense this is the most difficult case for a thermostat, but paradoxically, the assumption facilitates the mathematical analysis. We will assume throughout the following that $\mathrm{N_d} = \mathrm{N_c}$, i.e. that there are no constraints in the system. Our potential will therefore be assumed to be

$$U(\boldsymbol{q}) = \frac{1}{2}\boldsymbol{q}^T\boldsymbol{A}\boldsymbol{q},$$

for $\boldsymbol{A}$ a symmetric, positive definite $\mathrm{N_c} \times \mathrm{N_c}$ matrix, with

$$\nabla U(\boldsymbol{q}) = \boldsymbol{A}\boldsymbol{q}.$$

We will further assume $\boldsymbol{M} = \boldsymbol{I}$. If this is not the case, then one may introduce the rescaling $\tilde{\boldsymbol{q}} = \boldsymbol{M}^{-1/2}\boldsymbol{q}, \tilde{\boldsymbol{p}} = \boldsymbol{M}^{1/2}\boldsymbol{p}$ in order to rewrite the system in a form that does not involve $\boldsymbol{M}$. Thus the system under study is

$$\mathrm{d}\boldsymbol{q} = \boldsymbol{p}\mathrm{d}t, \tag{8.30}$$

$$\mathrm{d}\boldsymbol{p} = -\boldsymbol{A}\boldsymbol{q}\mathrm{d}t - \xi\boldsymbol{p}\mathrm{d}t, \tag{8.31}$$

$$\mathrm{d}\xi = \mu^{-1}[\|\boldsymbol{p}\|^2 - \mathrm{N_d}\mathrm{k_B}\mathrm{T}]\mathrm{d}t - \gamma\xi\mathrm{d}t + \sigma\mathrm{d}\mathrm{W}. \tag{8.32}$$

Let $\boldsymbol{u}_1, \boldsymbol{u}_2, \ldots, \boldsymbol{u}_{\mathrm{N_c}}$ be the eigenvectors of $\boldsymbol{A}$ with $\lambda_1, \ldots, \lambda_{\mathrm{N_c}}$ the corresponding eigenvalues. Observe that if $\boldsymbol{q}_0 \cdot \boldsymbol{u}_k = \boldsymbol{p}_0 \cdot \boldsymbol{u}_k = 0$, then, using (8.30)–(8.31), we have at this point

$$\boldsymbol{u}_k \cdot \boldsymbol{p} = 0, \qquad \boldsymbol{u}_k \cdot [-\boldsymbol{A}\boldsymbol{q} - \xi\boldsymbol{p}] = -\lambda_k \boldsymbol{u}_k \cdot \boldsymbol{q} = 0,$$

which implies that $V_k = \{(\boldsymbol{q}, \boldsymbol{p}) | \boldsymbol{u}_k \cdot \boldsymbol{q} = 0, \boldsymbol{u}_k \cdot \boldsymbol{p} = 0\}$ is an invariant submanifold. Therefore we must require that $\boldsymbol{u}_k \cdot \boldsymbol{q}$ and $\boldsymbol{u}_k \cdot \boldsymbol{p}$ do not simultaneously vanish, that is, $(\boldsymbol{q}, \boldsymbol{p}) \in \mathcal{D} = \mathbb{R}^{2\mathrm{N_c}} \setminus \bigcup_{k=1}^{\mathrm{N_c}} V_k$.

Theorem 8.1 *If A is symmetric positive definite and has distinct eigenvalues, then (8.30)–(8.32) satisfies the Hörmander condition on $\mathcal{D} \times \mathbb{R}$.*